

 VISTHAAPAN 

 AI +  GIS +  OR

For  Safe &  Efficient  Disaster Relocation

Final Complete Framework

VISTHAAPAN -

Vulnerability Intelligence and Spatial Transit for HAZard-Affected Population Allocation Network

VISTHAAPAN is an AI + GIS + Operations Research based decision-support platform that enables authorities to proactively identify vulnerable habitations, determine unsafe zones, evaluate safe relocation sites and their carrying capacity, and generate an optimal, cost-efficient, constraint-aware relocation plan.

The key idea is that VISTHAAPAN doesn't stop at "Who is at risk?" or "Where is safe?" It also solves:

> "Given limited capacity at safe locations, who should be moved where, how many people should be allocated to each site, and what is the most feasible/economical plan?"

1. Problem VISTHAAPAN Solves

Current disaster management is often heavily focused on response after a disaster.

VISTHAAPAN shifts the approach toward:

> Predict → Identify → Prioritize → Plan → Allocate → Relocate

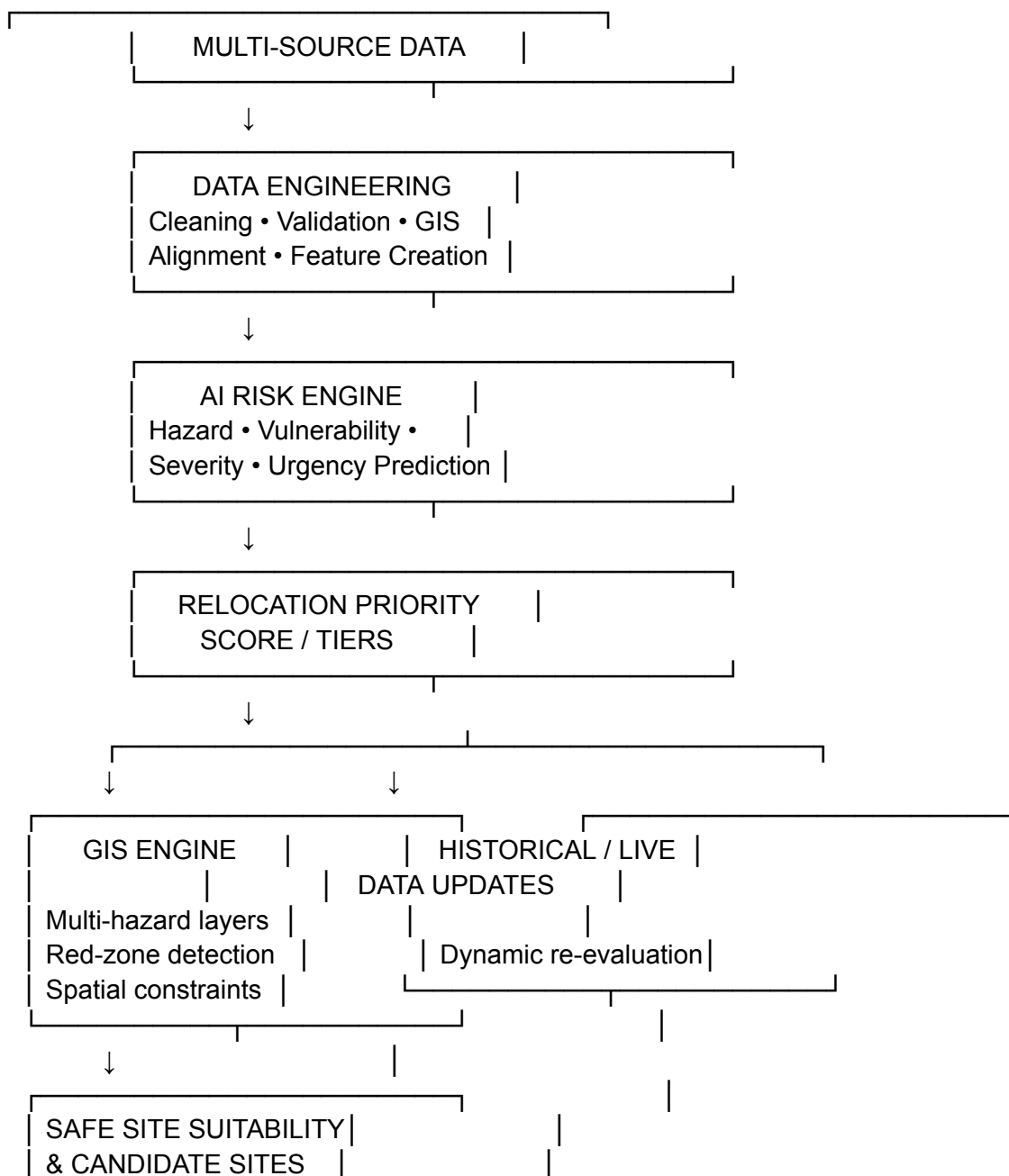
It addresses five connected problems:

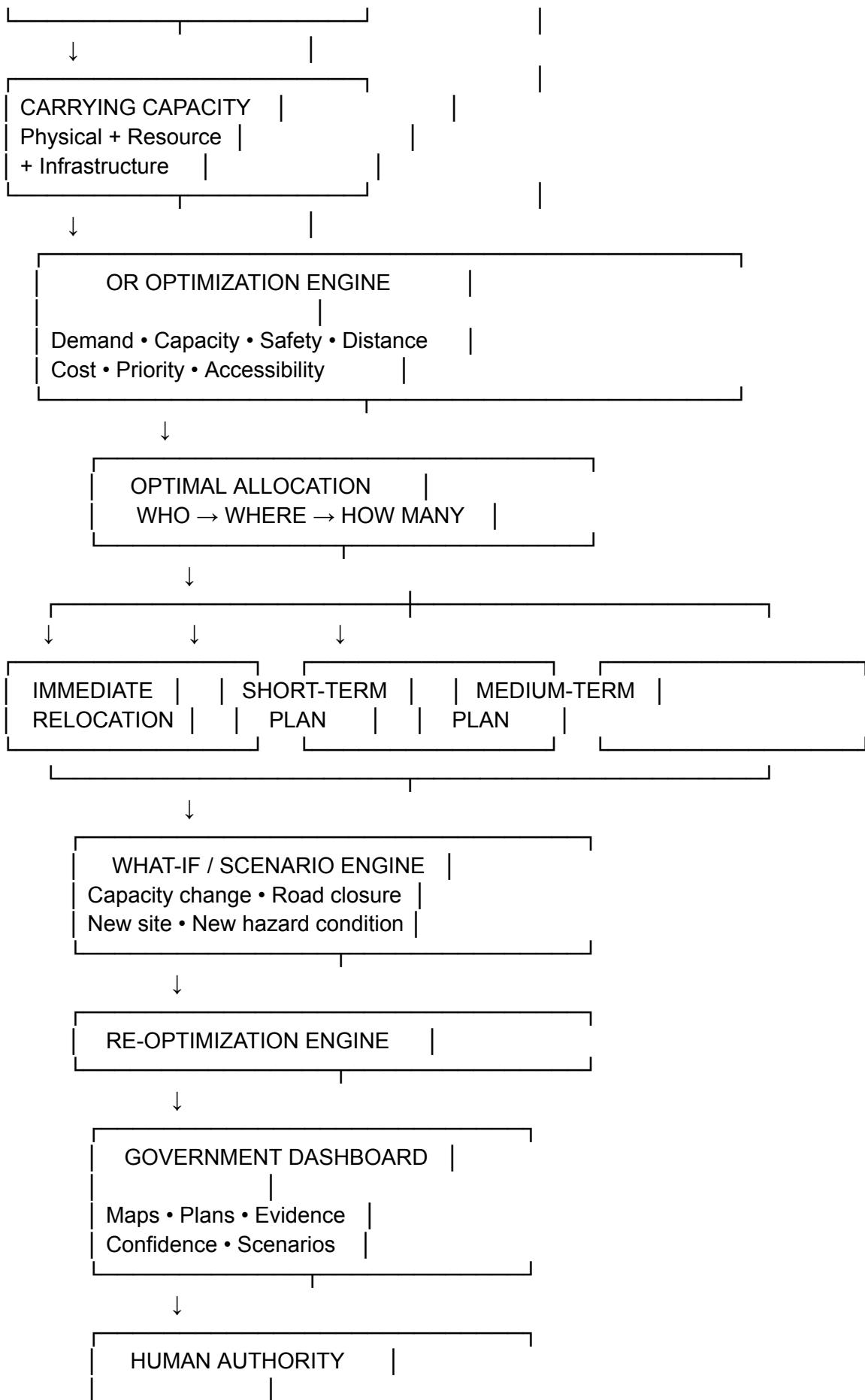
1. Which habitations are exposed to hazardous conditions?
2. Which vulnerable populations require relocation first?
3. Which areas are unsuitable for permanent habitation?

4. Which alternative sites are genuinely suitable and how many people can they safely accommodate?

5. How should limited safe-site capacity be optimally distributed among competing vulnerable habitations?

2. 🏗️ Overall Architecture





| Accept • Modify • Reject |

3. Multi-Source Data Layer

VISTHAAPAN combines multiple types of information.

A. Hazard Data

Depending on the region:

Flood

Landslide

Cloudburst / extreme rainfall

Coastal erosion

Earthquake

Lightning

Other relevant hazards

Along with:

Hazard intensity

Frequency

Historical events

Forecast information where available

Hazard-prone zones

B. Terrain & Geographic Data

Elevation

Slope

Drainage

River proximity

Land-use / land-cover

Soil

Terrain characteristics

C. Population & Vulnerability

Population

Households

Population density

Vulnerable population groups where reliable data is available

Critical facilities serving the population

Historical disaster impact

D. Infrastructure

Roads

Bridges

Hospitals

Schools

Water facilities

Electricity

Emergency facilities

Existing shelters

Public buildings

E. Historical Disaster Data

Used to understand:

Previous affected locations

Frequency

Severity

Repeatedly affected settlements

Historical evacuation/relocation patterns

4. 🛠️ Data Engineering Layer

Before AI or GIS processing:

Data validation

Check:

Missing values

Invalid coordinates

Duplicate records

Outliers

Inconsistent units

Different spatial resolutions

Spatial harmonization

Bring different datasets into a common:

Coordinate system

Spatial resolution

Geographic boundary

Feature engineering

Generate meaningful features such as:

Distance from river

Mean/maximum slope

Historical disaster frequency

Population density

Distance to hospital

Road accessibility

Hazard exposure

Infrastructure availability

5. 🤖 AI Risk & Vulnerability Engine

AI analyses the collected information to estimate the risk associated with each habitation.

The model can consider:

> Hazard Exposure + Historical Severity + Population Vulnerability + Infrastructure Exposure
+ Terrain

Possible models:

Logistic Regression — baseline

Random Forest

XGBoost / CatBoost

The output isn't simply:

> Safe / Unsafe

Instead, it can generate:

Risk Score

For example:

> Village A → 0.92 risk

Vulnerability Score

> Village A → 0.86 vulnerability

Urgency / Severity

> Village A → High

These are then combined into a Relocation Priority Weight (RPW).

6. Relocation Priority Engine

Each habitation receives a priority value.

For example:

Habitation	Risk	Vulnerability		Priority Tier
Village A	0.94	0.89	0.92	Immediate
Village B	0.81	0.84	0.82	Immediate
Village C	0.65	0.61	0.63	Short-term
Village D	0.42	0.48	0.45	Medium-term

The exact formula/weights would be determined through validation rather than arbitrarily claiming that one universal formula is scientifically correct.

Important:

AI determines priority.

It does not directly decide the final relocation destination.

7. 🌐 GIS Engine

GIS converts the AI output into spatial decision-making.

Multi-hazard mapping

Overlay relevant:

Flood zones

Landslide zones

Erosion zones

Rainfall/hazard layers

Terrain

Population

Infrastructure

Red-Zone identification

Areas meeting defined hazard/safety criteria are identified as:

> ● Unsuitable for permanent habitation

The system can dynamically update these zones when new data becomes available.

Critical principle

Don't blindly combine different hazards into one weighted number.

Some conditions should be hard constraints.

For example:

> If a candidate site is inside a prohibited high-hazard zone → Site unavailable

8. 🏠 Safe-Site Suitability Engine

VISTHAAPAN identifies potential relocation sites outside unsuitable areas.

Each candidate site can be evaluated using:

Safety

Distance from affected habitation

Road accessibility

Healthcare access

Water availability

Sanitation

Electricity

Existing infrastructure

Terrain suitability

Land availability

Transportation feasibility

The system produces:

> Candidate Site Suitability Score

while keeping critical safety restrictions as hard constraints.

9. 🏠 Carrying Capacity Assessment

This is one of the most important components.

A site shouldn't be considered capable of accommodating people simply because it has enough physical land.

Effective capacity can be constrained by:

Physical area

Water

Sanitation

Shelter

Healthcare

Electricity

Roads/access

Other essential infrastructure

Conceptually:

> Effective Capacity = minimum of the relevant limiting capacities

For example:

Physical capacity = 15,000

Water capacity = 12,000

Shelter capacity = 10,000

Healthcare capacity = 13,000

Effective capacity = 10,000

This creates a realistic constraint for the optimization engine.

10. 🚚 OR-Based Population Allocation

This is where VISTHAAPAN goes beyond simple “rank the villages” or “find the nearest shelter.”

Imagine:

Village A → 8,000 people

Village B → 7,000 people

↓

Safe Site X

Capacity = 10,000

Both villages need relocation, but Site X cannot accommodate everyone.

So:

> Who gets the capacity?

This becomes an Operations Research allocation problem.

Decision variable

Let:

> x_{ij} = number of people relocated from habitation i to site j

Constraints

Demand

People from a habitation must either be allocated or explicitly identified as unmet demand.

> $\sum_j x_{ij} + u_i = D_i$

where u_i is unmet demand.

Capacity

> $\sum_i x_{ij} \leq C_j$

A site can never be assigned more people than its effective capacity.

Safety

Unsafe allocations:

$$x_i > 0$$

Non-negativity

$$x_i \geq 0$$

11. 💰 Economic Optimization

The OR engine considers:

Relocation distance

Transportation cost

Site suitability

Population priority

Available capacity

Accessibility

Other operational constraints

Conceptually:

> Maximize priority-weighted suitability while minimizing relocation cost/burden

For example:

High priority + highly suitable + low relocation cost

should generally be preferred over:

Low priority + poorly suitable + expensive relocation

The AI's probability is not simply treated as a transportation cost. It is transformed into a priority/benefit factor used appropriately within the optimization formulation.

12. ⚠ Capacity Deficit Handling

This is critical.

Suppose:

Total population requiring relocation = 30,000

Total safe available capacity = 22,000

The system must not pretend that everyone can be accommodated.

Instead:

> ● Capacity Deficit: 8,000 people

Then VISTHAAPAN can identify:

Remaining unmet population

Additional candidate sites

Temporary alternatives

Future capacity requirements

This makes the system much more realistic.

13. 🕒 Phased Relocation Planning

The final allocation can be divided into:

● Immediate

Highest-priority populations requiring urgent action.

● Short-Term

Populations requiring relocation after immediate needs are addressed and additional capacity/infrastructure becomes available.

● Medium-Term

Permanent relocation and rehabilitation planning.

This directly addresses the PS's requirement for different relocation horizons.

14. Dynamic Re-optimization

Disaster conditions aren't static.

If:

Hazard intensity changes

A road becomes unavailable

Site capacity decreases

A new safe site is added

Infrastructure becomes unavailable

Population estimates change

VISTHAAPAN can:

> Update → Recalculate → Re-optimize

without rebuilding the entire plan manually.

15. What-If Scenario Engine

This can become a major demo feature.

The officer can simulate:

Scenario A

> “Site X capacity decreases by 30%.”

→ Recalculate.

Scenario B

> “Road to Site X is blocked.”

→ GIS removes the feasible connection.

→ OR recalculates.

Scenario C

> “New Site Y becomes available with 8,000 capacity.”

→ Add site.

→ Re-optimize.

The dashboard then shows:

> Previous allocation vs New allocation

This demonstrates actual decision-support capability.

16. Explainable Decision Support

Every major recommendation should have an explanation.

For example:

Why is Village A Immediate Priority?

> Priority Score: 0.92 — HIGH

Contributing factors:

High flood exposure

High population vulnerability

Repeated historical flooding

Critical infrastructure exposure

Limited local safe capacity

Why Site X?

> Suitability: HIGH

Reasons:

Outside Red Zone

Adequate capacity

Good road access

Nearby healthcare

Adequate water availability

Lower relocation cost

This makes the system auditable and understandable.

17. Data Confidence Layer

Every recommendation should also communicate the reliability of its inputs.

Example:

Risk Score : 0.92
Confidence : HIGH
Data Freshness : 3 days
Data Coverage : 94%
Historical Data : Available

If data quality is poor:

>  Low-confidence recommendation

This is especially important because disaster decisions can have serious consequences.

18. Human-in-the-Loop

VISTHAAPAN should explicitly remain a:

> Decision-Support System

—not an autonomous authority.

The officer receives:

System Recommendation

↓

Accept / Modify / Reject

If modified:

> Reason for override

can be recorded.

For example:

System:

Village A → Site X

Officer:

Village A → Site Y

Reason:

Local infrastructure at Site X currently unavailable.

This allows domain expertise and information not present in datasets to remain part of the final decision.

19. Government Dashboard

The final interface should bring everything together.

Main map

Hazard layers

Red Zones

Vulnerable habitations

Candidate relocation sites

Routes/connections

Population

Priority levels

Habitation panel

Village A

Population: 8,000

Risk: 0.94
Vulnerability: 0.89
Priority: 0.92
Relocation Tier: IMMEDIATE

Site panel

Site X

Effective Capacity: 10,000
Suitability: HIGH
Water: Adequate
Healthcare: 4.2 km
Accessibility: GOOD

Allocation panel

Village A → Site X : 6,000
Village A → Site Y : 2,000
Village B → Site X : 4,000
Village B → Site Z : 3,000

System status

People requiring relocation : 15,000
Available safe capacity : 17,000
Unmet demand : 0

20.  Validation & Reliability

VISTHAAPAN should validate each layer independently.

AI

Precision

Recall

F1

ROC-AUC

Calibration/Brier score

GIS

Compare identified zones against:

Authoritative hazard maps

Historical disaster events

Known affected regions

Carrying Capacity

Validate against:

Known administrative capacities

Infrastructure/resource limits

Applicable standards where available

OR

Automatically verify:

No capacity violations

No unsafe allocations

All constraints satisfied

Correct handling of insufficient capacity

Correct response to scenario changes

21.  Complete End-to-End Example

Suppose:

Village A → 8,000 people

Village B → 7,000 people

AI identifies:

A → Priority 0.94

B → Priority 0.81

GIS determines:

Site X → Safe, capacity 10,000

Site Y → Safe, capacity 4,000

Site Z → Safe, capacity 5,000

OR receives:

Demand

Capacity

Priority

Distance

Transportation cost

Safety constraints

Accessibility

Site suitability

and generates:

A → X : 7,000

A → Y : 1,000

B → X : 3,000

B → Y : 3,000

B → Z : 1,000

Then an officer says:

> “Road to Site X is unavailable.”

VISTHAAPAN:

GIS:

X connection removed

↓

OR:

Re-optimize

↓

New allocation generated

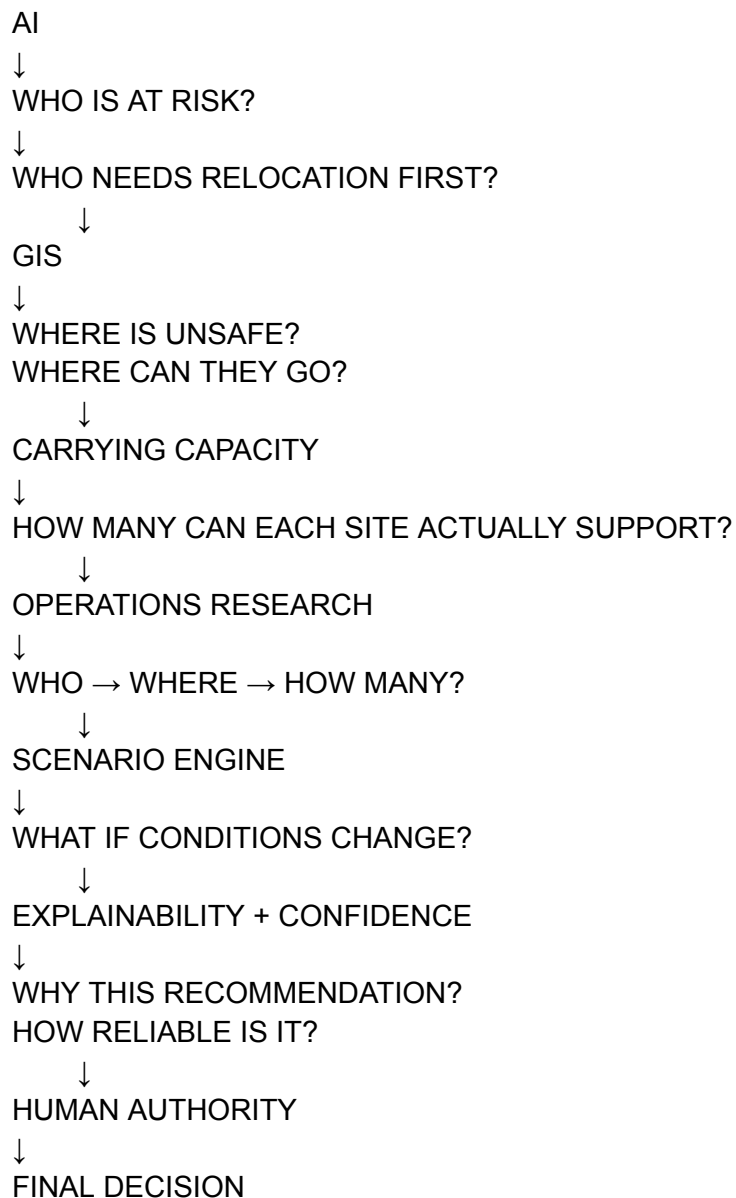
The dashboard shows:

> Previous Plan vs Revised Plan

along with the new capacity deficit, if any.

22. 🌱 The Core Intelligence of VISTHAAPAN

The entire system can be summarized as:



The Fundamental Innovation

The innovation shouldn't be presented as:

> “We invented a new AI algorithm.”

That's not what makes VISTHAAPAN strong.

The innovation is the problem formulation and integration:

> VISTHAAPAN integrates AI-driven risk and vulnerability assessment, GIS-based spatial feasibility, carrying-capacity constraints, and Operations Research-based population allocation into a unified proactive disaster-relocation decision-support system.

Existing systems can tell authorities about hazards.

GIS systems can identify suitable locations.

AI can predict risk.

Optimization can solve allocation.

But VISTHAAPAN connects these into one decision chain:

Risk → Priority → Safety → Capacity → Allocation → Cost → Action

And that is the part I would make the central story of the SIH presentation.

Final VISTHAAPAN Framework

VISTHAAPAN

Vulnerability Intelligence and Spatial Transit for Hazard-Affected Population Allocation Network

> An AI-GIS-OR powered proactive disaster relocation decision-support platform that identifies vulnerable habitations, dynamically maps unsafe zones, evaluates safe-site

carrying capacity, prioritizes populations for relocation, and optimally allocates limited relocation capacity while considering safety, accessibility, distance and economic cost.

Technology Stack

AI/ML → Risk, vulnerability, severity, priority

GIS → Spatial analysis, Red Zones, safe sites, accessibility

Capacity Modeling → Realistic site limits

Operations Research → Optimal population allocation

Scenario Engine → What-if analysis & re-optimization

Explainability → Why this decision?

Confidence Layer → How reliable?

Human-in-the-loop → Authority makes the final decision

Dashboard → Actionable government decision support

And the philosophy:

> Don't just predict disasters.

Don't just map hazards.

Don't just identify safe places.

Don't just rank vulnerable populations.

Decide how limited safe resources should be optimally used to relocate the people who need it most. 🤖